MCC Entrance Test, August 2026 — Day 2 — Solutions

Level T4

Test Information. Level T4 — 180 minutes.
6 problems: 2 Easy (10 pts), 2 Medium (15 pts, two parts), 2 Hard (25 pts, three parts); 100 points total.
Areas: Algebra (×2), Combinatorics, Geometry, Number Theory (×2).
Rules: show your work (full, justified arguments earn full credit at this level); no calculators; read all six before choosing where to start; box your final answers.
A past MCC entrance test, free to try: https://nghia71.github.io/readiness

Problem 1 (10 pts).[1] Consider the numbers 11111, 111111, and 122222 (note that 122222 = 11111+111111). Raising these to the 4th power directly produces numbers too large for an ordinary calculator to evaluate exactly. Find, without a calculator,

$$\sqrt{\frac{11111^4 + 111111^4 + 122222^4}{2}}.$$

Official solution. Let $a = 11111$, $b = 111111$, so $122222 = a + b$. Write $u = a^2 + ab + b^2$ and $v = ab$; a quick check shows

$$u + v = a^2 + 2ab + b^2 = (a + b)^2, \qquad u - v = a^2 + b^2.$$

Since $a^4 + b^4 = (a^2 + b^2)^2 - 2(ab)^2 = (u - v)^2 - 2v^2$ and $(a + b)^4 = (u + v)^2$, adding these gives

$$a^4 + b^4 + (a + b)^4 = \left[(u - v)^2 - 2v^2\right] + (u + v)^2 = \underbrace{u^2 - 2uv - v^2}_{(u-v)^2-2v^2} + \underbrace{u^2 + 2uv + v^2}_{(u+v)^2} = 2u^2,$$

since the uv terms cancel and the v^2 terms cancel. Hence

$$\sqrt{\frac{a^4 + b^4 + (a + b)^4}{2}} = \sqrt{u^2} = u = a^2 + ab + b^2$$

(taking the nonnegative root, since $a, b > 0$) – with no need to expand $(a + b)^4$, find any 4th power directly, or take a square root of a 21-digit number (all genuinely out of reach for a calculator here). Substituting $a = 11111$, $b = 111111$ – and noting along the way that $a^2 = 123454321$ and $b^2 = 12345654321$ are the repunit's classic "counting up, counting down" palindromes:

$$a^2 + ab + b^2 = 123454321 + 1234554321 + 12345654321 = \boxed{13703662963}.$$

□

Problem 2 (10 pts).[2] Let x and y be non-negative integers. Find all pairs (x, y) for which $x + 3^y$ and $y + 3^x$ are *consecutive* integers (that is, they differ by exactly 1, in either order).

Official solution. Let $A = x + 3^y$ and $B = y + 3^x$; we want $|A - B| = 1$, where

$$A - B = (x - y) - (3^x - 3^y).$$

Swapping x and y swaps A and B (hence flips the sign of $A - B$), so by this symmetry it suffices to find all solutions with $x \geq y$, and then also record their swaps. Write $k = x - y \geq 0$.

Case $k = 0$ $(x = y)$: $A - B = 0 - (3^x - 3^x) = 0$, not ± 1. No solution.

Case $k = 1$ $(x = y + 1)$: $3^x - 3^y = 3^y(3 - 1) = 2 \cdot 3^y$, so $A - B = 1 - 2 \cdot 3^y$. For $y = 0$: $A - B = 1 - 2 = -1$, so $|A - B| = 1$ – a solution, $(x, y) = (1, 0)$. For $y \geq 1$: $2 \cdot 3^y \geq 6$, so $A - B \leq 1 - 6 = -5$, giving $|A - B| \geq 5$.

Case $k \geq 2$: $3^x - 3^y = 3^y(3^k - 1) \geq 3^k - 1$ (using $3^y \geq 1$), and $3^k - 1 \geq 3^2 - 1 = 8$ for $k \geq 2$. The quantity $(3^k - 1) - k$ equals 6 at $k = 2$ and strictly increases in k (since $\left(3^{k+1} - 1 - (k + 1)\right) - (3^k - 1 - k) = 2 \cdot 3^k - 1 > 0$), so $(3^k - 1) - k \geq 6$ for all $k \geq 2$. Hence

$$|A - B| = 3^y(3^k - 1) - k \geq (3^k - 1) - k \geq 6 > 1, \text{ so no solution for } k \geq 2.$$

So the only solution with $x \geq y$ is $(x, y) = (1, 0)$; by the symmetry noted above, the only solution with $x \leq y$ is its swap $(x, y) = (0, 1)$. So $(x, y) \in \boxed{\{(0, 1),\ (1, 0)\}}$. □

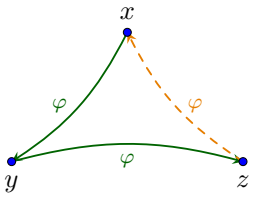
[1] Japan Mathematical Olympiad, Preliminary Round. [2] Romanian Mathematical Olympiad, County Round (Etapa Judeţeană).

Problem 3 (15 pts).[3]

Define the function

$$\varphi(r) = r^3 - 4r^2 - 16r + 60.$$

Starting from an integer x, let $y = \varphi(x)$ and $z = \varphi(y)$; say the triple (x, y, z) *closes* if $\varphi(z) = x$ once again.



(a) Find all real numbers r with $\varphi(r) = r$. [5 pts]

(b) Determine all integer triples (x, y, z) for which the loop closes as described. [10 pts]

Official solution. **Part (a).** $\varphi(r) - r = r^3 - 4r^2 - 17r + 60$. *If integer r is a root of φ, r is a divisor of* 60. Testing shows that $r = -4, 3$, and 5 are roots, thus $r^3-4r^2-17r+60 = (r-3)(r^2-r-20) = (r-3)(r-5)(r+4)$. So $\varphi(r) = r$ exactly when $(r + 4)(r - 3)(r - 5) = 0$, i.e. $r \in \boxed{\{-4,\ 3,\ 5\}}$.

Part (b). Step 1 – outside $[-4, 5]$, φ escapes. Let $g(r) = \varphi(r) - r = (r + 4)(r - 3)(r - 5)$.

If $r > 5$: all three factors of $g(r)$ are positive, so $g(r) > 0$, i.e. $\varphi(r) > r$. Since $\varphi(r) > r > 5$ too, the exact same reasoning applies again to $\varphi(r)$, and again after that – so once a value exceeds 5, repeatedly applying φ produces a strictly increasing sequence that never returns.

If $r < -4$: all three factors are negative, so $g(r) < 0$, i.e. $\varphi(r) < r < -4$; by the same reasoning, repeated application produces a strictly decreasing sequence that stays below -4 forever.

Step 2 – so x, y, z can't be outside $[-4, 5]$ either. The triple cycles $x \to y \to z \to x$, so for any one of x, y, z – call it t – applying φ three times in a row (once per step of the cycle) must land back on t itself.

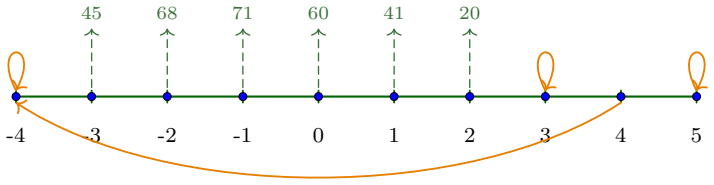
If $t > 5$, Step 1 gives $t < \varphi(t) < \varphi(\varphi(t)) < \varphi(\varphi(\varphi(t)))$ – a strict increase after three steps, which can never equal t again. If $t < -4$, the same idea gives a strict decrease instead. Both are contradictions, so every one of x, y, z must lie in $[-4, 5]$ – hence, being integers, in $\{-4, -3, -2, -1, 0, 1, 2, 3, 4, 5\}$.

Step 3 – narrow the ten candidates down. Computing φ at each of these ten integers:

r	-4	-3	-2	-1	0	1	2	3	4	5
$\varphi(r)$	-4	45	68	71	60	41	20	3	-4	5

By Step 2, $y = \varphi(x)$ must also land in $[-4, 5]$ – it's one of the three cycling values too. For $x \in \{-3, -2, -1, 0, 1, 2\}$, the table shows $\varphi(x)$ already lands outside $[-4, 5]$, so none of these six values can even start a closing loop. That leaves only $x \in \{-4, 3, 4, 5\}$ to check.

The picture below sums up Step 3: orange arrows mark the four values that stay inside $[-4, 5]$ (self-loops at the fixed points $-4, 3, 5$, and 4 landing on -4), while green arrows mark the six values that escape at once, taking themselves out of the running.



Step 4 – check each survivor directly. For $x = -4, 3, 5$: these are exactly the fixed points from part (a), so $y = z = x$ and $\varphi(z) = x$ holds trivially. For $x = 4$: $y = \varphi(4) = -4$, $z = \varphi(-4) = -4$, and $\varphi(z) = \varphi(-4) = -4 \neq 4$ – the loop does *not* close. So the complete solution set is $(x, y, z) \in \boxed{\{(-4, -4, -4),\ (3, 3, 3),\ (5, 5, 5)\}}$. □

[3] Bundeswettbewerb Mathematik (Federal Mathematics Competition, Germany).

Problem 4 (15 pts).[4] Consider a row of 19 slots, each containing one of two symbol types, labeled 1 and 2. Call such a sequence *balanced* if, among its 16 windows of 4 consecutive slots, no two windows show the same pattern of types.

(a) Prove that a balanced sequence exists. [6 pts]

(b) Prove that in every balanced sequence, the types in the first three slots match, in order, the types in the last three slots. [9 pts]

Official solution. **Part (a).** Consider the sequence 2 2 2 2 1 1 1 1 2 2 1 2 1 1 2 1 2 2 2.

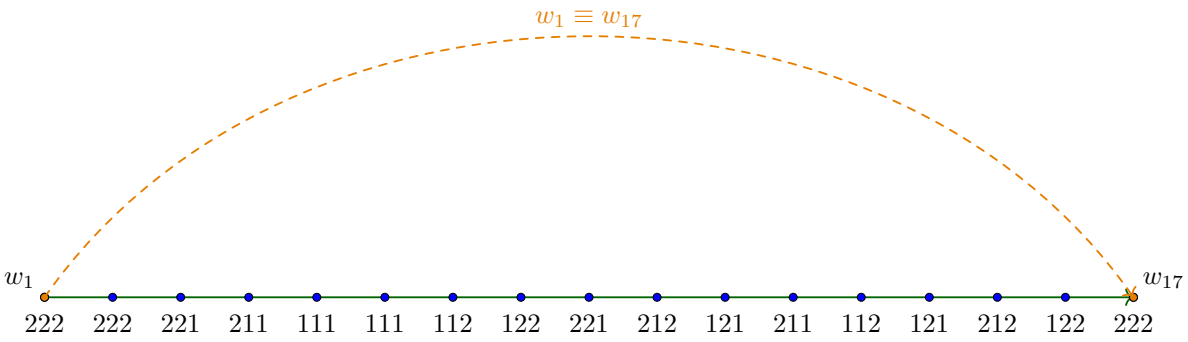
There are 16 windows of 4 consecutive slots (starting at positions 1 through 16), and there are exactly $2^4 = 16$ possible length-4 patterns over $\{1, 2\}$. Listing the 16 windows of this sequence in order gives

$$2222, 2221, 2211, 2111, 1111, 1112, 1122, 1221, 2212, 2121, 1211, 2112, 1121, 1212, 2122, 1222,$$

which is a full list of all 16 possible patterns with none repeated. So this sequence is balanced, proving existence.

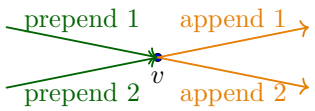
Part (b). Step 1 – turn this into a walk on a small graph. Treat *each of the* 17 *windows of* 3 *consecutive slots (positions* 1 *through* 17*) as a* **node** – one of the $2^3 = 8$ possible 3-symbol strings over $\{1, 2\}$ – and *each of the* 16 *windows of* 4 *consecutive slots as a directed* **edge**, from its first three symbols to its last three symbols (append one symbol, drop the first). Consecutive 3-windows are joined by exactly the edge given by the 4-window straddling them, so the sequence of nodes $w_1, w_2, \ldots, w_{17}$ traces a walk, one edge per step.

Since the sequence is balanced, all 16 possible 4-symbol patterns appear among the 16 windows, each exactly once – and each pattern is exactly one edge of the graph – so this walk uses *every edge exactly once.* Here is what that walk looks like for the specific sequence from part (a):



Part (b) claims that $w_1 = w_{17}$ always happens, not just in this example – that's what the rest of this proof shows in general.

Step 2 – every node has exactly 2 ways in and 2 ways out. Any 3-symbol string v has exactly 2 outgoing edges (append a 1 or a 2) and exactly 2 incoming edges (prepend a 1 or a 2): So every node has in-degree = out-degree = 2, counting among all 16 edges.



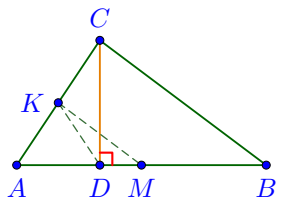
Step 3 – only the start and end node can be unbalanced. Consider any node v that is *not* the start w_1 and *not* the end w_{17}. Every visit the walk makes to v both arrives (one in-edge) and departs (one out-edge), pairing up v's edges one-to-one as they get used. Since the walk eventually uses *every* edge at v (Step 1), and v has exactly 2 in-edges and 2 out-edges (Step 2), this pairs up perfectly – no contradiction.

But suppose $v = w_1$ and $v \neq w_{17}$. The walk's very first step leaves v with no preceding arrival to pair it with, so once all of v's edges are used, departures from v exceed arrivals by exactly 1: but both totals equal 2,

[4] Bulgarian National Mathematical Olympiad, Regional Round.

so this would require $2 = 2 + 1$ – impossible. Symmetrically, if $v = w_{17}$ and $v \neq w_1$, arrivals would have to exceed departures by 1, again giving $2 = 2 + 1$, impossible.

So it cannot be that $w_1 \neq w_{17}$; we must have $w_1 = w_{17}$, i.e. the first three slots' types equal the last three slots' types. □



Problem 5 (25 pts).[5] In triangle ABC with $\angle A \geq \angle B$, point M is the midpoint of side AB, and point D is the foot of the altitude from C to line AB.

(a) Prove that if $\angle A = 2\angle B$, then $AC = 2\,MD$. [10 pts]

(b) Let K be the midpoint of AC. Prove that triangle KDM is isosceles with $KD = DM$ if and only if $\angle A = 2\angle B$. [4 pts]

(c) Prove the converse: if $AC = 2\,MD$, then $\angle A = 2\angle B$. [11 pts]

Official solution. **Part (a).** Let K be the midpoint of AC. Since $\angle ADC = 90°$, DK is the median to the hypotenuse of right triangle ADC, so $KD = KA = KC = \frac{1}{2}AC$, so $\triangle AKD$ isosceles with $\angle ADK = \angle DAK = \angle A$.

Since M, K are the midpoints of AB, AC, segment MK is a midline of $\triangle ABC$, so $MK \parallel BC$; as AB is a transversal cutting these parallel lines, corresponding angles give $\angle KMD = \angle ABC = \angle B$.

Since A, D, M are collinear on line AB, $\angle ADK$ is an exterior angle of triangle KDM at vertex D, so

$$\angle ADK = \angle DKM + \angle KMD \implies \angle A = \angle DKM + \angle B \implies \angle DKM = \angle A - \angle B = \angle B = \angle KMD.$$

Thus $\triangle KDM$ is isosceles, so $DM = KD$. Since $KD = \frac{1}{2}AC$, $AC = 2\,KD = \boxed{2\,MD}$.

Part (b). The two relations derived in part (a) – $\angle KMD = \angle B$ (midline/corresponding-angles fact) and $\angle DKM = \angle A - \angle B$ (exterior-angle fact) – hold for *any* $\triangle ABC$, independent of whether $\angle A = 2\angle B$.

$\triangle KDM$ is isosceles with $KD = DM$ exactly when the angles opposite these sides are equal:

$$KD = DM \iff \angle DKM = \angle KMD \iff \angle A - \angle B = \angle B \iff \boxed{\angle A = 2\angle B}.$$

Part (c). Since K gives $KD = \frac{1}{2}AC$, the hypothesis $AC = 2\,MD$ gives $KD = \frac{1}{2}AC = MD$, so $\triangle KDM$ is isosceles with $KD = DM$. By part (b), $KD = DM$ forces $\angle A - \angle B = \angle B$, i.e. $\angle A = \boxed{2\angle B}$.

Remark. The hypothesis $\angle A \geq \angle B$ is essential here: without it, $AC = 2\,MD$ admits a second, spurious family of triangles with $\angle B = 90° + \frac{1}{2}\angle A$ (so $\angle B$ obtuse) and $\angle A \neq 2\angle B$ – e.g. $\angle A = 20°$, $\angle B = 100°$, $\angle C = 60°$ satisfies $AC = 2\,MD$ yet $\angle A \neq 2\angle B$. This spurious family always has $\angle A < \angle B$, so the hypothesis $\angle A \geq \angle B$ rules it out, restoring the converse above. □

Problem 6 (25 pts).[6] Let x, y be integers and let p be a prime number satisfying

$$x^2 - 3xy + p^2y^2 = 12p.$$

(a) Show that if $y = 0$, then $p = 3$, and find the two resulting values of x. [5 pts]

(b) Show that if $y \neq 0$, then p must satisfy $4p^2 - 48p - 9 \leq 0$, and conclude that only $p \in \{2, 3, 5, 7, 11\}$ need be checked. [6 pts]

(c) Complete the analysis, and find all triples (x, y, p). [14 pts]

[5] Sharygin Geometry Olympiad. [6] Turkey Junior National Mathematical Olympiad.

Official solution. **Part (a).** Setting $y = 0$: $x^2 = 12p$. Since $12p = 4 \cdot 3p$ and 4 is already a perfect square, $12p$ is a perfect square exactly when $3p$ is. As p is prime, if $p \neq 3$ then 3 and p are distinct primes each appearing to the first power in $3p$, which cannot be a perfect square; so $p = 3$. Then $x = \boxed{\pm 6}$, $\quad p = \boxed{3}$.

Part (b). View the equation as a quadratic in x: $x^2 - (3y)x + (p^2y^2 - 12p) = 0$. For a real (let alone integer) solution x to exist, its discriminant must be nonnegative:

$$\Delta = 9y^2 - 4(p^2y^2 - 12p) = y^2(9 - 4p^2) + 48p \geq 0 \implies y^2(4p^2 - 9) \leq 48p.$$

Since $y \neq 0$ is an integer, $y^2 \geq 1$, so a necessary condition is $4p^2 - 9 \leq 48p \iff \boxed{4p^2 - 48p - 9 \leq 0}$.

The roots of $4p^2 - 48p - 9 = 0$ are $p = \frac{48 \pm \sqrt{48^2 + 4 \cdot 4 \cdot 9}}{8} = \frac{48 \pm \sqrt{2448}}{8}$, and $\sqrt{2448} \approx 49.48$, so the inequality holds for $p \lesssim \frac{48 + 49.48}{8} \approx 12.19$. Since $13 > 12.19$, thus $p < 13$, leaving only $\boxed{p \in \{2, 3, 5, 7, 11\}}$ to check.

Part (c). For each surviving prime, $y^2 \leq \dfrac{48p}{4p^2 - 9}$ bounds $|y|$, and $\Delta = y^2(9 - 4p^2) + 48p$ must be a perfect square.

Case $p = 2$: $4p^2 - 9 = 7$, so $y^2 \leq \frac{96}{7} \approx 13.7$, giving $|y| \leq 3$. $\Delta = -7y^2 + 96$: for $y = \pm 1, \pm 2, \pm 3$, $\Delta = 89, 68, 33$ respectively – none are perfect squares.

Case $p = 3$: $4p^2 - 9 = 27$, so $y^2 \leq \frac{144}{27} \approx 5.3$, giving $|y| \leq 2$. $\Delta = -27y^2 + 144$: for $y = \pm 1$, $\Delta = 117$, not a perfect square; for $y = \pm 2$, $\Delta = 36 = 6^2$, a perfect square. Solving $x = \dfrac{3y \pm 6}{2}$:

$$y = 2 \implies x = 6 \text{ or } 0; \qquad y = -2 \implies x = 0 \text{ or } -6.$$

Each of $(6, 2), (0, 2), (0, -2), (-6, -2)$ (with $p = 3$) checks directly against the original equation, e.g. $6^2 - 3(6)(2) + 9(2)^2 = 36 - 36 + 36 = 36 = 12(3)$.

Case $p = 5$: $4p^2 - 9 = 91$, so $y^2 \leq \frac{240}{91} \approx 2.6$, giving $|y| \leq 1$. $\Delta = -91y^2 + 240 = 149$ for $y = \pm 1$ – not a perfect square ($12^2 = 144$, $13^2 = 169$).

Case $p = 7$: $4p^2 - 9 = 187$, so $y^2 \leq \frac{336}{187} \approx 1.8$, giving $|y| \leq 1$. $\Delta = -187y^2 + 336 = 149$ for $y = \pm 1$ – again not a perfect square.

Case $p = 11$: $4p^2 - 9 = 475$, so $y^2 \leq \frac{528}{475} \approx 1.1$, giving $|y| \leq 1$. $\Delta = -475y^2 + 528 = 53$ for $y = \pm 1$ – not a perfect square ($7^2 = 49$, $8^2 = 64$).

Combining with part (a)'s $y = 0$ solutions $(6, 0, 3)$ and $(-6, 0, 3)$, the complete solution set is

$$(x, y, p) \in \boxed{\{(6, 0, 3),\ (-6, 0, 3),\ (6, 2, 3),\ (0, 2, 3),\ (0, -2, 3),\ (-6, -2, 3)\}}.$$

□